Portable Telehealth System for Refugee Camps

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Problem Statement

Refugee and displacement camps lack a centralized and reliable healthcare record system, leading to fragmented medical histories, repeated treatments, and life-threatening errors such as unknown allergies. Frequent movement of patients, rotating NGO healthcare providers, and limited or no internet connectivity make it difficult to maintain continuous and accessible medical records. This results in poor healthcare delivery, lack of specialist access, and ineffective disease monitoring across camps.

Real-World Impact

The proposed system has the potential to significantly improve healthcare delivery in refugee and displacement camps by ensuring continuity, safety, and accessibility of medical information. By enabling portable and secure health records, it reduces medical errors, enhances diagnosis accuracy, and ensures timely treatment even in resource-constrained environments.

Our Solution

- *We propose a Portable, Offline-First Digital Health Identity System that securely stores and manages patient medical records, ensuring continuous healthcare access across refugee camps. The system is designed to work in low or no internet environments and enables healthcare providers to access, update, and share patient data seamlessly.*

Key Features

- **Identity & Access**
- **Unique Health ID** (Biometric + QR-based)
- **No document required** for identification
- **Role-based access control** (Doctor / Nurse / Admin)

- **Offline-First System**
- Works **without internet connectivity**
- Stores data in **local encrypted database**
- Ensures **no data loss in remote areas**

- **Smart Data Sync**
- Supports **intermittent connectivity syncing**
- Automatic **cloud backup when online**
- **Conflict resolution** using timestamps

- **Telehealth Support**
- **Low-bandwidth audio/video consultation**
- Enables **remote specialist access**
- “Store & forward” consultation option

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- **Medical Records Management**
- Tracks **patient history, vitals, and treatments**
- Maintains **medication history & allergy alerts**
- Supports **immunization tracking**

- **Data Portability**
- Patient data accessible **across multiple camps**
- **QR scan / offline transfer (Bluetooth)**
- Ensures **continuity of care**

- **Family & Community Features**
- **Family linking system** (children under one ID)
- **Anonymized health reports** for camp management

- **Security & Ethics**
- **End-to-end encryption**
- **No raw biometric storage**
- **Patient consent-based access**

Identity Logic

In refugee and displacement camps, individuals often lack official identification documents, making it extremely difficult to maintain consistent medical records. To address this challenge, our system introduces a robust, multi-layered identity logic that ensures every patient can be uniquely identified, authenticated, and linked to their medical history across different locations and time periods.

Offline Sync Logic

- Our system follows an “offline-first” approach, where all patient data is stored and updated locally on the device, and synchronized with the cloud only when internet connectivity becomes available. This ensures uninterrupted healthcare services even in completely disconnected environments.

Telehealth Feature

- Supports low-bandwidth audio/video consultation
- Uses compressed communication for weak networks
- Enables store & forward consultation (send data when internet is available)
- Allows remote specialist access from anywhere

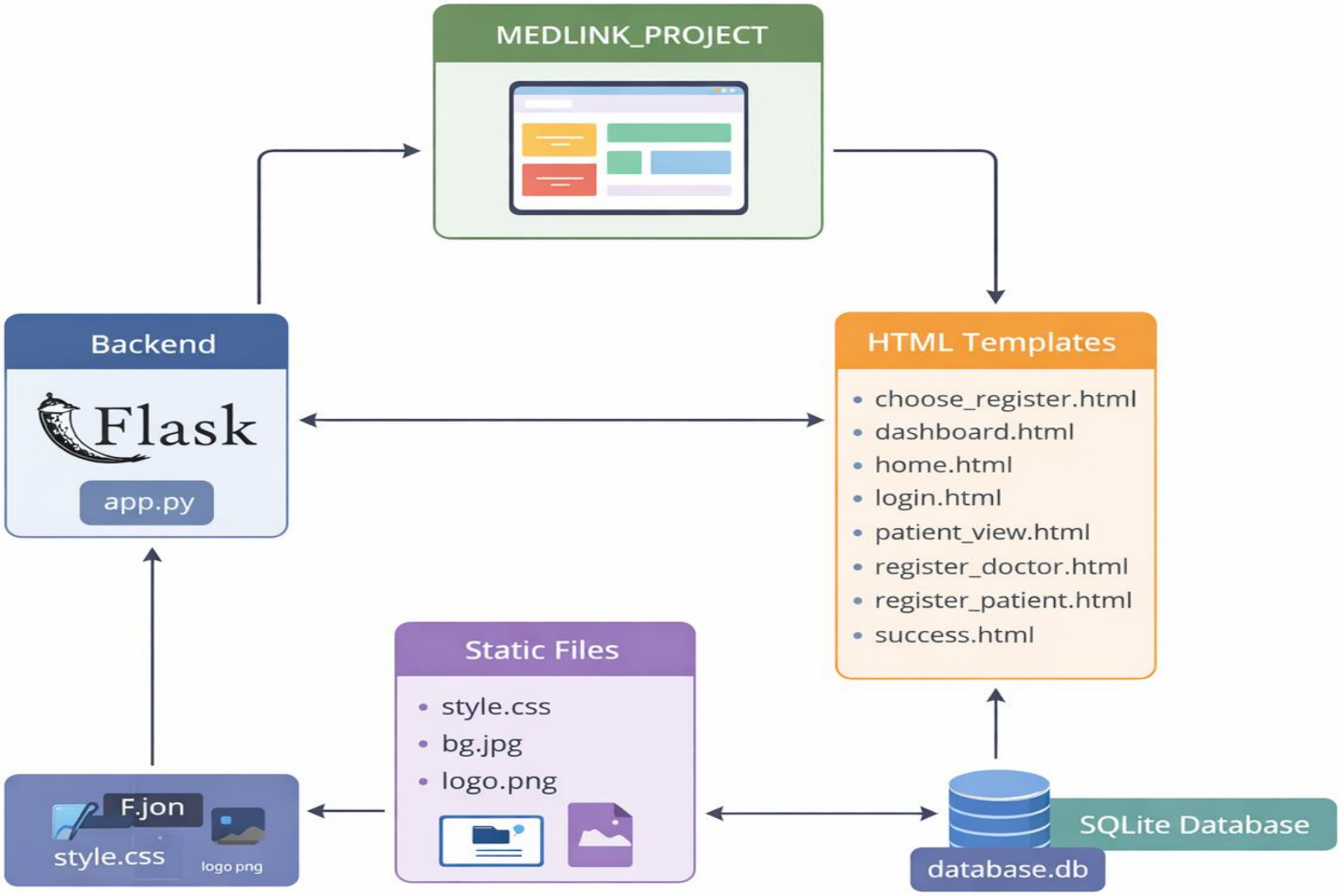
Data Portability

- Patient records can be accessed using QR scan anywhere
- Supports Bluetooth / offline file transfer when no internet
- Enables secure data movement between camps
- Ensures continuous medical history across locations

Security & Ethics

- All data is **encrypted end-to-end**
- **No raw biometric data stored** (only encrypted hash)
- **Role-based access control** (Doctor / Nurse / Admin)
- Access only with **patient consent**
- Designed to protect **privacy of vulnerable populations**

System Architecture Diagram



Tech Stack

- Frontend: React / Flutter
- Backend: Node.js
- Database: SQLite (offline storage)
- Cloud: Firebase (backup & sync)
- AI (optional): Python for matching & suggestions

Future Scope

- Integration with **wearable health devices**
- AI-based **disease prediction system**
- Collaboration with **global NGOs and health organizations**
- Development of a **multi-camp unified health network**
- Real-time **epidemic tracking system**

Conclusion

- Our project provides a **portable telehealth system for refugee camps** that ensures continuous healthcare even without internet or documents. It uses a **unique biometric and QR-based identity system** to track patients securely.
- The system works in an **offline-first mode**, allowing doctors to access and update records anytime, and automatically syncs data when connectivity is available. It also supports **remote teleconsultation and secure data sharing** between camps.

THANK YOU!